

Chi-Square Tests, Pt 2

CSU, Chico Math 351

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Introduction

We can extend the goodness-of-fit test to deal with the case in which the null hypothesis is that the distribution of the data belongs to a particular family. The alternative hypothesis is that the data have a distribution that is **not** a member of the family.

Though we might suspect that the data come from, say, the exponential distribution, we have no prior information to suggest the values of the parameters of the distribution (i.e. $\hat{\lambda}$).

Introduction

We therefore have to estimate the parameters from the data. The difference between this test and the test in the previous section is that the test statistic D has probabilities that are based on the parametric family. Also, the degrees of freedom are reduced by the **number of parameters** being estimated.

The test statistic is a modified version of D that we learned in the previous section.

Definition

In order to carry out a χ^2 test of goodness-of-fit of the hypotheses, the test statistic D must be modified because the expected number of np_{i_0} of observations of type i in a random sample of n observations is no longer completely specified by the null hypothesis H_0 .

The modification that is used is to replace np_{i_0} by the M.L.E. of this expected number under the assumption that H_0 is true. If we let $\hat{\theta}$ denote the parameter θ based on the observed counts $N_1, N_2, \dots, N_k$ for the k possible outcomes, then the statistic D is defined as follows:

$$D = \sum_{i=1}^k \frac{[N_i - n\pi_i(\hat{\theta})]^2}{n\pi_i(\hat{\theta})}$$

Definition

Just like in the previous section we reject the H_0 when D is larger than some critical value. The degrees of freedom are adjusted though since we are estimating parameters from the data. Thus, D will still follow a χ^2 distribution; however, D will have $t - 1 - s$ degrees of freedom since we are estimating s parameters $\theta_1, \dots, \theta_s$.

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Example 1 (from Larsen & Marx)

Over a 3-year period (1096 days), the London *Times* reported daily numbers of death notices for women over the age of eighty. Test the hypothesis that these observations form a random sample from a **Poisson distribution**.

Number deaths, i	Obs. Freq. k_i	Est.Exp.Freq. $n\hat{p}_{io}$
0	162	126.8
1	267	273.5
2	271	294.9
3	185	212.1
4	111	114.3
5	61	49.3
6	27	17.8
7	8	5.5
8	3	1.4
9	1	0.3
10+	0	0.1

Example 1 (from Larsen & Marx)

Where did the expected values come from? The Poisson model states:

$$P(i) = e^{-\lambda} \lambda^i / (i!), i = 0, 1, 2, \dots$$

And though we don't know λ , its Maximum Likelihood Estimator (from earlier) is simply $\bar{X}$, total number of fatalities over total number of days. In this case, $\hat{\lambda} = 2.157$, and we can calculate $E(X = i)$ for each number of fatalities. In RStudio, we can do this using:

$$E(X=i) = n\hat{p}_{io} = n * \text{dpois}(x=i, \text{lambda}=2.157)$$

Since the $n\hat{p}_{io} \geq 5$ requirement is violated for larger values of i , we can reconstruct the table so there is instead a “7+” category.

Example 1 (from Larsen & Marx)

Number deaths, i	Obs. Freq. k_i	Est.Exp.Freq. $n\hat{p}_{io}$
0	162	126.8
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3	185	212.1
4	111	114.3
5	61	49.3
6	27	17.8
7+	12	7.3

From the above, the test statistic d_1 will be:

$$d_1 = \frac{(162 - 126.8)^2}{126.8} + \dots + \frac{(12 - 7.3)^2}{7.3} \approx 25.91$$

And degrees of freedom will be:

$$(\text{number of groups} - 1 - \text{number of parameters}) = 8 - 1 - 1 = 6.$$

Example 1 (from Larsen & Marx)

If using R, the resulting p-value would be:

$$1 - \text{pchisq}(25.91, 6) \approx 0.0002$$

Or we can compute the critical chi-square value above which we reject H_0 :

$$\text{qchisq}(1 - \alpha, 6) \approx 12.592$$

From the above, this is more than sufficient to reject the null hypothesis. This data does not appear to follow a Poisson distribution, as the counts we observe diverges too much from the counts we expect.

Additional R Resources

Chapter 10.4 of the Speegle & Clair text gives a useful look at other ways to organize your data to perform a χ^2 Goodness of Fit test:

<https://probstatsdata.com/tabular.html>

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Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Richard J Larsen, Morris L Marx. *An Introduction to Mathematical Statistics and Its Applications*. Prentice Hall, fifth edition, 2012.

Adapted from notes by Dr. Gray.